CS23532 – COMPUTER NETWORKS – LAB MANUAL

Practical – 2

Aim: Study of different types of Network Cables.

Objective: Understand different types of network cables, their characteristics, advantages, disadvantages and applications.

Cable Type	Category / Type	Maximum Data Transmission	Advantages	Disadvantages	Application / Use
UTP	Category 3 Category 5 Category 5e	10 Mbps Up to 100 Mbps 1 Gbps	Cheaper in cost Easy to install Smaller diameter	More prone to EMI and noise	10Base-T Ethernet Fast Ethernet Gigabit Ethernet
STP	Category 6, 6a	Up to 10 Gbps	Shielded Faster than UTP Less susceptible to noise/interference	Expensive Greater installation effort	Gigabit Ethernet 10G Ethernet (55 m) Data centres
SSFP	Requested type	Not specified in supplied manual	Additional shielding/foil can improve protection from interference	More expensive and harder to terminate than basic UTP	Where enhanced shielding is required
Coaxial Cable	RG-6 RG-59 RG-11	10–100 Mbps	High bandwidth Immune to interference Low-loss bandwidth Versatile	Limited distance Cost Bulky size	Television network High-speed internet connections Signal transmission up to about 500 m (manual)
Fibre Optic Cable	Single mode Multi mode	100 Gbps	High speed High bandwidth High security Long distance	Expensive Requires skilled installers Manual states around 100 m	High-speed networking Long-distance communication

Types covered: UTP, STP, SSFP, Coaxial Cable and Fibre Optic Cable.

Reference note: The supplied lab manual lists SSTP (Category 7) rather than SSFP. SSFP is therefore included here because it was specifically requested, and its row is not presented as content taken from the manual.

1. UTP – Unshielded Twisted Pair Cable

UTP uses pairs of insulated copper wires twisted together. The twisting helps reduce interference between pairs.

Advantages	Disadvantages	Applications / Uses
• Cheaper in cost • Easy to install • Smaller overall diameter	• More prone to electromagnetic interference (EMI) and noise	• 10Base-T Ethernet • Fast Ethernet • Gigabit Ethernet

2. STP – Shielded Twisted Pair Cable

STP is a twisted-pair cable with shielding that provides better protection from noise and interference.

Advantages	Disadvantages	Applications / Uses
• Shielded • Faster than UTP in the referenced comparison • Less susceptible to noise and interference	• Expensive • Greater installation effort	• Gigabit Ethernet • 10G Ethernet (55 m) • Widely used in data centres

3. SSFP – Shielded/Foiled Twisted-Pair Cable

SSFP is included as requested. The supplied manual does not provide a dedicated SSFP specification, so this section is intentionally limited.

Advantages	Disadvantages	Applications / Uses
• Additional shielding/foil can improve protection from interference • Useful in electrically noisy environments	• Higher cost • More care may be required during installation	• Networks requiring enhanced shielding

4. Coaxial Cable

Coaxial cable has a central conductor, insulation, metallic shielding and an outer protective jacket. The manual identifies RG-6, RG-59 and RG-11.

Advantages	Disadvantages	Applications / Uses
• High bandwidth • Immune to interference • Low-loss bandwidth • Versatile	• Limited distance • Cost • Bulky size	• Television networks • High-speed internet connections

5. Fibre Optic Cable

Fibre optic cable carries information using light through optical fibre. The manual identifies single-mode and multi-mode fibre.

Advantages	Disadvantages	Applications / Uses
• High speed • High bandwidth • High security • Long distance	• Expensive • Requires skilled installers	• High-speed networking • Long-distance communication • Backbone links

Making Your Own Ethernet Cross-Over / Straight Cable

Tools and parts needed:

• Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well over shorter distances.

• A crimping tool for inserting pins and stripping/cutting cable shielding.

• Two RJ45 plugs.

• Optional two plug shields.

Difference between crossover cable and straight cable

A straight-through cable uses the same wiring arrangement at both ends. A crossover cable uses different wiring arrangements at the two ends, using the A/B diagrams in the supplied manual.

Step 1: Thread shields onto the cable.

Step 2: Strip approximately 1.5 cm of cable shielding from both ends.

Step 3: Untangle the four twisted pairs and arrange the wires according to the required A/B diagram.

Step 4: Align the wires evenly and insert them into the RJ45 plug while maintaining their order.

Step 5: Push the cable fully into the plug; the notch should be just over the cable shielding.

Step 6: Insert the plug into the crimping tool and press down carefully.

Step 7: Repeat for the other end using diagram B for a crossover cable or diagram A for a straight-through cable.

Testing: Connect the cable and attempt to establish communication between the intended devices.

Observation / Viva Questions

1. What is the difference between a cross cable and a straight cable?

2. Which type of cable is used to connect two PCs — straight or cross cable?

3. Which type of cable is used to connect a router/switch to your PC — straight or cross cable?

4. Find out the category of twisted-pair cable used in your lab to connect the PC to the network socket.

5. Write down your understanding, challenges faced and output received while making a twisted-pair cross/straight cable.

Student Observation

Parameter	Observation / Result
Cable used	_____
Cable category	_____
Cable type	_____
Connectivity result	_____
Challenges faced	_____ _____

Result: The characteristics and applications of UTP, STP, SSFP, coaxial and fibre-optic cables were studied. The construction procedure for Ethernet straight-through and crossover cables was also reviewed.